Answering the judges in plain words

RadZero, Graviton 2026, Problem Statement 31. This sheet is for every member of the team. No formula on it. If you can say the five lines below, you can take any question on the floor.

If you remember one sentence: we check what the satellite's AI is about to do against what the satellite can physically do, before it happens, and we repair the damage in under a millisecond without switching anything off.

Say this in thirty seconds

Read it out loud twice before you walk in. It is about ninety words, so it fits in half a minute at a normal pace.

> "Our project is called RadZero. A small satellite uses a normal computer chip to steer itself, and in space, radiation can flip one bit inside the numbers that chip is using. Nothing crashes, but the chip now gives a wrong steering command, and the satellite can start spinning. Today's fix is to switch the computer off and restart it. That takes minutes to hours, and the satellite is out of control the whole time. We added a guard that checks every steering command against what the satellite can physically do, before the command goes out. If the numbers look wrong, we put a clean copy back in about a millisecond, and the computer never restarts."

The words you will hear, in one line each

If a judge uses one of these words, answer with the plain version in the right-hand column. Using the simple version makes you sound confident, not ignorant.

Word a judge may use	What it means	What you should say
Bit flip	One bit inside the chip changes by accident.	"One wrong number."
Single-event upset	The technical name for a bit flip caused by radiation.	"A radiation hit."
Weights	The numbers the AI learned from data.	"The AI's settings."
Torque	A twisting force that turns something.	"The steering command."
Reaction wheel	A spinning wheel that turns the satellite.	"The wheels that turn it."
Checksum or CRC	A fingerprint of a block of data.	"A fingerprint, so we can tell it changed."
Watchdog	A timer that restarts a computer if it stops responding.	"A dead-man's switch."
Latch-up	A short circuit inside the chip that needs the power cut.	"The chip shorts out."
Radiation-hardened	A chip built to survive radiation, and much slower.	"A space chip, far slower."
Readiness level	How close a technology is to flying.	"How far it is from orbit."
Telemetry	The readings a satellite sends to the ground.	"The readings we send down."

The twelve questions you will actually get

Learn the answer, not the wording. If a judge asks something not on this list, use the closest answer and stop talking. The last line under each answer is for the follow-up, when they want more.

1. What does your project actually do?

We sit between the satellite's AI and its motors. Before any steering command goes out, we check that the numbers are sane and that the command is physically possible for that satellite. If either check fails, we write a clean copy of the AI's settings back over the damaged ones and re-run the same step. The satellite keeps steering the whole time.

If they push: it is a safety guard, not a new AI. We did not train a better model, we built a check around one.

2. Why does the AI break in the first place?

Up there nothing shields a normal chip, so high-energy particles from the Sun and from deep space hit it all the time. When one hits a memory cell it can flip a bit, which changes a number the AI is using. The AI has no way to know the number changed, so it keeps working and gives a wrong answer.

If they push: this is a known, measured problem in space electronics, not a hypothetical one.

3. How do you spot the damage?

Two ways. Inside the step: are the numbers sane, and is the command physically possible. In the background: we keep a fingerprint of the clean copy and re-check one quarter of the AI's numbers every cycle, so all of it is checked about every four cycles.

If they push: the fast checks catch about half the faults. The background scan catches the other half, and we say so on our slides.

4. How do you repair it?

We keep two clean copies of the AI's settings in ordinary memory. When a check fails, we write the clean numbers back over the damaged ones while the program is still running, then check that same step again before anything reaches the motors.

If they push: no restart, no reload, no reboot. The program is the same program, start to finish.

5. How fast is that, and is it fast enough?

The satellite makes a new decision ten times a second, so it has one hundred milliseconds for each one. Our average repair is 0.68 milliseconds, and the worst case in fifty trials was 3.3 milliseconds. That leaves more than ninety-six percent of the time budget spare.

If they push: we are roughly a hundred times faster than the deadline needs.

6. Does the satellite stop working while you repair?

No. The program never restarts and the control loop never misses a step. That is the whole point. A restart means minutes with no steering, and a satellite that is already spinning may never recover its pointing.

If they push: we measured it. Fifty faults, fifty repairs, zero restarts.

7. How do you know it works?

We wrote a tool that flips bits on purpose, in every way radiation does, and ran fifty trials end to end with the satellite's motion carried from one step to the next. All fifty were caught and repaired. We also have fifty-two automated tests, and anyone can re-run both and get the same result on their own machine.

If they push: the numbers come from a file in our repository, not from a slide we drew.

8. Is this not what memory error-correction already does?

Memory correction fixes a bad bit when the computer reads it. It cannot tell you whether the answer the AI produced makes sense. We check the answer itself against physics, so we catch problems that memory correction cannot see, including ones that never touched the AI's settings at all.

If they push: we check the decision, not only the storage.

9. What if the damage hits your own checker?

That is the honest gap, and it is a fair question. Our checker's own settings sit in ordinary memory, so a hit there could quietly blind it while it still looks healthy. The fix is a small separate chip that keeps a copy of those settings and asks us to prove they have not changed. That chip is grams and milliwatts, not a second computer.

If they push: it is in our design and in our limits section, not yet in the code. Do not pretend otherwise.

10. What about damage that physically burns the chip?

Software cannot fix that. A particle can short the chip internally, and the only cure is to cut its power. We designed a small hardware circuit that notices the current spike in under ten microseconds and cuts power before the chip burns, then brings it back. That part is a design on paper, not something we built.

If they push: we already say this in our limits. Do not claim we can recover a burnt chip.

11. Why not just use a chip built for space?

Space-rated chips are roughly a hundred times slower and far more expensive, and they still fail. Small satellites are moving to ordinary phone-class chips because that is the only way to get enough computing on a small power budget. Our approach lets them keep the cheap fast chip without losing control of it.

If they push: we spend software instead of money, weight and power.

12. What is not finished?

Four things, and we say all four before you ask. We inject the faults in software, so no radiation beam has been fired at it. We test one flipped bit at a time, not bursts. Our headline numbers come from a normal processor, not the graphics chip. And it has never flown. That is why we call it lab-validated.

If they push: saying this first is exactly why you should believe the rest of our numbers.

The five numbers, and what each one proves

Number	What it proves	Why it matters
0.68 ms	How long an average repair takes.	Speed
50 of 50	Faults caught and repaired in our trial.	Reliability
3.3 ms	The slowest single repair we saw.	Worst case
96.7%	Share of the hundred-millisecond budget left over.	Headroom
52	Automated tests, all passing.	Discipline

If a judge asks for a number that is not here, do not invent one. Say: I do not have that in front of me, it is in our benchmark file, and I will send it to you. That answer costs you nothing. A made-up number costs you the round.

Five things not to say

- **We stop radiation damage.** Say instead: We cannot stop the hit. We detect and repair the damage it does.
- **It never needs a reboot.** Say instead: Not for this kind of fault. A burnt chip or a frozen computer still needs one.
- **It is ready for space.** Say instead: It is lab-validated, readiness level four. Say that and it sounds stronger.
- **We tested it on the GPU.** Say instead: Our published numbers come from the processor. Say which one you mean.
- **We tested it with real radiation.** Say instead: We simulated the faults in software. A beam test is our next step.

Who answers what

Split it before you go in, so four people do not answer the same question at once. Whoever is asked answers. Everyone else stays quiet and nods. A team that talks over itself looks like it has not rehearsed.

- **The story.** Why this matters and what we built. Opens and closes.
- **How it works.** The four steps and the live demo. Takes anything about the checks.
- **The numbers.** The trial, the tests, the timing. Takes anything with a figure in it.
- **The limits.** What is not finished and what comes next. Takes every hard question.

If you are stuck

Say this, slowly: **"I don't have that number in front of me. It's in our benchmark file, and I'll send it to you."** Then stop. Do not fill the silence with a guess. Every number in this project is in one file in the repository, so that answer is always true, and it always sounds better than a wrong figure.